Pricing the Western Gateway, Fairly

A congestion charge for the Etobicoke–Gardiner corridor, and the conditions under which it could actually be built

Briefing note to senior decision-makers | UTTAN 2026 Student Competition

Bahar Shahkaram, Siddhartha Pahari, Jainish Solanki | University of Toronto | September 2026

Executive Summary

Congestion costs the GTHA more than $44 billion a year, and Toronto has already tried to price it: Council approved tolls on the Gardiner and DVP in 2016, and the Province refused. The binding problem is not whether a charge would work. It is whether one can be authorised, survive a vote, and be worth having once the alternatives are counted honestly.

We modelled a peak-period charge on the Gardiner's western gateway, Highway 427 to the Humber River, as a three-level game: an authority sets a price, travellers respond, and voters and governments decide whether it proceeds. Three findings should change how the corridor is approached.

First, **coverage matters more than price**. A charge on the expressway alone pushes 11.6% more traffic onto The Queensway; charging the full screenline removes the free bypass and turns diversion negative.

Second, **a charge is not the strongest delay instrument available**. A no-charge operational package removes 45.2% of peak delay against 39.4% for the recommended charge. The two together reach 64.4%. Pricing earns its place on revenue, durability and demand management, not on delay.

Third, **sequencing is worth more than the price**. At an identical charge, the difference between the best and worst order of operations is 0.56 welfare points – larger than anything the rate itself buys.

Recommendations

1. **Charge the full western screenline, not the expressway.** Price every inbound crossing between Highway 427 and the Humber, including Lake Shore Boulevard West and The Queensway. A freeway-only charge at $4 breaches a 10% local-diversion cap on City streets; screenline coverage reverses it.

2. **Set the rate at $10–12 for a 06:30–09:30 window, with income and transit-poor credits.** Solving price and start year together puts the charge at $12. Above that it fails our equity floor, which is the binding constraint – not public acceptability, which barely moves with the rate.
3. **Deliver the operational package regardless of the charge.** Adaptive signals, ramp management, transit priority, parking pricing and GO frequency remove more peak delay on their own than the charge does, and carry none of its political cost. They are complements, not rivals.
4. **Sequence deliberately: transit first, then a time-limited pilot, then a vote.** This path reaches 0.434 expected discounted welfare against -0.130 for charging immediately. None of that is better policy; all of it is a better chance the policy exists.
5. **Resolve the 2027 Gardiner transfer before committing to a date.** The Province's refusal in 2017 implies it values the political cost above $3.91 per charged trip. Above roughly $3.50, waiting until after the transfer dominates rushing a deal before it.

Problem Statement

The Gardiner's western approach carries roughly 120,000 two-way weekday trips across a screenline where the expressway, an arterial and a secondary arterial run parallel into the same downstream bottleneck. It is the strongest candidate in the GTHA for a first charge: against 6,000 randomly weighted criterion sets, Etobicoke–Gardiner ranked first in 95.8% of them. It also scores worst of five candidates on governance, which is the problem.

The charged segment is 8.2 km of an 18.5 km trip. That geometry, not the price, drives the central risk: the cheapest way to avoid a gateway charge is not to stop driving but to leave the expressway before the gantry, run an arterial, and re-enter past it. Any design that ignores this understates diversion onto residential streets.

Toronto's 2016–17 episode is the second constraint. Council voted for tolls; the Province refused and offered a gas-tax transfer instead. A corridor charge therefore has to clear two independent gates – a public that will tolerate it and two governments that can agree – and the Gardiner's transfer to the Province in late 2027 moves the second one.

Methodology

We built a bounded-corridor Stackelberg model. Five links are solved to a stochastic user equilibrium by the Method of Successive Averages: travellers choose the lowest perceived generalized cost, but the cost of driving depends on how many others drive.

Travel choice is **nested logit** across eight actions – pay and stay, two arterial bypasses, carpool, retime, GO rail, TTC, and forgo the trip. Multinomial logit was rejected deliberately: under its independence assumption a driver priced off the expressway is as likely to board a train as to divert onto Lake Shore, which would understate the study's central risk. Demand is segmented into 70 groups (seven sub-areas × five income quintiles × schedule flexibility).

Link performance is fitted on three links and **held out on two**, whose predicted-versus-observed errors (−6.3%, −3.0%) are a genuine out-of-sample check. Every input carries a provenance tier – published, derived or estimated – and the most consequential estimated inputs are swept in a tornado analysis and a Monte Carlo.

Nine operating and rejected schemes were used as **evidence rather than illustration**, including three places where the record contradicts our model. We report those contradictions rather than resolving them in our favour.

Analysis

The charge is not the best delay instrument. Table 1 compares the options on the quantity most often quoted.

Table 1: Peak delay removed, against a common no-charge baseline

Option	Delay removed
\$5 screenline charge alone	35.5%
Recommended \$10 charge with credits	39.4%
No-charge operational package	**45.2%**
Charge *and* operational package	**64.4%**

A charge only outperforms good operations if it is priced hard and compensated sparingly, which is the opposite of an equitable design. The defensible position is that both should be done, and that pricing is justified by revenue, durability and demand management rather than by delay.

Equity is the binding constraint, and the design that fixes it is what makes the scheme passable. A flat charge is regressive: the lowest income quintile bears 4.7 times the burden of the highest as a share of income. Targeted credits cut that to 2.2 times, at the cost of revenue and some congestion relief. Against published benchmarks (Table 2) even the credited design remains more regressive than every operating charge – a fact we report rather than round away.

Table 2: Suits index: 0 is proportional, negative is regressive

Scheme	Suits index
Stockholm / Helsinki	−0.09
Gothenburg	−0.13
Lyon (proposed)	−0.16
This corridor, recommended design	**−0.16**
This corridor, flat $10 charge	−0.20

The burden is local, not regional. Contrary to the New Jersey pattern often cited from New York, the 905 pays 36.2% of the charge while making 40.0% of trips – it is a net beneficiary, because Lakeshore West GO offers a genuine alternative. The heaviest burden falls on inner Etobicoke: Alderwood makes 6.5% of trips and pays 8.6% of the charge. Income explains only 46% of the variation in burden across sub-areas, which is why geographic equity needs treating separately from income equity.

Sequencing is worth more than the price. The same $10 charge produces very different expected outcomes depending on what precedes it (Table 3). The gap between the best ordering and charging immediately is 0.56 welfare points – larger than the difference between any two rates we tested. None of it is better policy; all of it is a better chance the policy exists, and the pilot does most of the work because it replaces a forecast with an experience.

Table 3: Expected discounted welfare over ten years, at an identical $10 charge

Path	Precedent	Support at gate	E[welfare]
Transit, pilot, then referendum	Stockholm	73%	**0.434**
Transit first, then charge	Stockholm, no pilot	52%	0.171
Transit, ride-hail surcharge, cordon	New York 2019–25	52%	0.120
Charge immediately	New York 2008	38%	−0.130
Operations, never charge	Toronto CMP	–	−0.197

Authorisation is a second, separate gate. A charge needs a public that will tolerate it *and* two governments that can agree. Today neither the City nor the Province can price the Gardiner alone, so they hold a mutual veto. After the corridor transfers in late 2027 the Province owns the road and authorises itself – and gains the option of a freeway-only charge, the very design that pushes traffic onto City streets. Because the political cost of authorising is unobservable, we invert it rather than assume it: the zone of agreement closes at $3.91 per charged trip before the transfer and $8.75 after. The Province did refuse in 2017, which places its true valuation above the first figure. On that reading, waiting until after the transfer dominates rushing a deal before it, and public support makes agreement materially easier – at $3.50 per charged trip, allowing the two gates to move together makes waiting 5.7 times more valuable.

The scheme is financially marginal. At $10 with credits, net revenue is roughly $5.6 million a year. Fixed collection costs do not scale down, so a single-corridor pilot can spend more collecting than it raises. Revenue is a reason to price a network, not this corridor alone.

What we are least sure of. The screenline demand is derived rather than counted, and observed travel times are estimates. The diversion finding depends on an assumed 55% of bypassing drivers rejoining the expressway; it reverses below 49%. Our demand response is also 2.1 points stronger than New York observed in its first year, which we record as unresolved rather than explained away. A published screenline count and probe travel times would settle more than any further modelling.

Use of AI Tools

AI tools were used to support research only, as permitted. Specifically: to assist in writing and debugging the Python simulation code underlying the analysis; to help structure the review of comparator schemes; and to check numerical results for internal consistency. All modelling assumptions, source selection, interpretation and every word of this briefing note are the authors' own. No AI tool was used to prepare this note or the accompanying presentation.

References

1. CBC News (2025, January 29). *Congestion costs the GTHA more than $44 billion a year.*
2. Eliasson, J. (2016). *Is Congestion Pricing Fair? Consumer and Citizen Perspectives on Equity Effects.* ITF Discussion Paper 2016-13.
3. Federal Highway Administration. *Income-Based Equity Impacts of Congestion Pricing: A Primer.* FHWA-HOP-08-040.
4. Greenlining Institute (2020). *Mobility Pricing: An Equity-First Approach.*
5. Shaheen, S., et al. (2019). *Mobility and the Sharing Economy: STEPS to Transportation Equity.* UC Berkeley TSRC.
6. San Francisco County Transportation Authority (2020). *Congestion Pricing Case Studies: London, Stockholm, New York.*
7. Regional Plan Association (2019). *Congestion Pricing in New York City.*
8. City of Toronto. *Gardiner Expressway and Don Valley Parkway: Tolling Options* (2016) and *Provincial Upload Agreement* (2024).
9. Metrolinx. *Business Case Guidance* and *Lakeshore West Corridor Service Plans.*
10. Transportation Tomorrow Survey (2022). Data Management Group, University of Toronto.

Appendix A – Model Structure

Five links: `gardiner_west` (the charged segment, 8.2 km), `gardiner_east` (shared downstream, 10.3 km), plus `lakeshore_west`, `lakeshore_east` and `queensway_west`. Link performance follows $t(v) = t_f + d_s + t_f\,\alpha(v/c)^\beta$, with β held at a literature value and α fitted on three links and transferred to two.

Appendix B – Selected Results

Recommended design: $10, full western screenline, 06:30–09:30, with income and transit-poor credits.

Indicator	**Value**
Peak vehicles	−7.6%
Peak delay removed	39.4%
Travel time saved	4.5 minutes
Net revenue	$5.6M per year
Travellers better off	63.1%
Support, before / after operation	38% / 61%
Pigouvian benchmark, charged segment	$9.73
Pigouvian, untolled downstream section	$18.34
Price of anarchy	1.14 ($55.9M per year)

Appendix C – Where the Record Contradicts This Study

1. **London.** Its traffic reduction was permanent; its delay reduction was not. Our 39.4% is a first-year figure. Background growth alone erodes it to 34.7% over ten years, and reproducing London's five-year reversion requires about 28% of freed capacity to be reallocated to buses and public realm – at which point traffic is down further still.
2. **Edinburgh.** Rejected 74.4% to 25.6% on *trust*, not economics: one in four believed the revenue would reach transport.
3. **Manchester.** A £3bn package was attached and the referendum was still lost. A credible package cannot rescue a charge nobody believes will work.